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VGG-NET Architecture

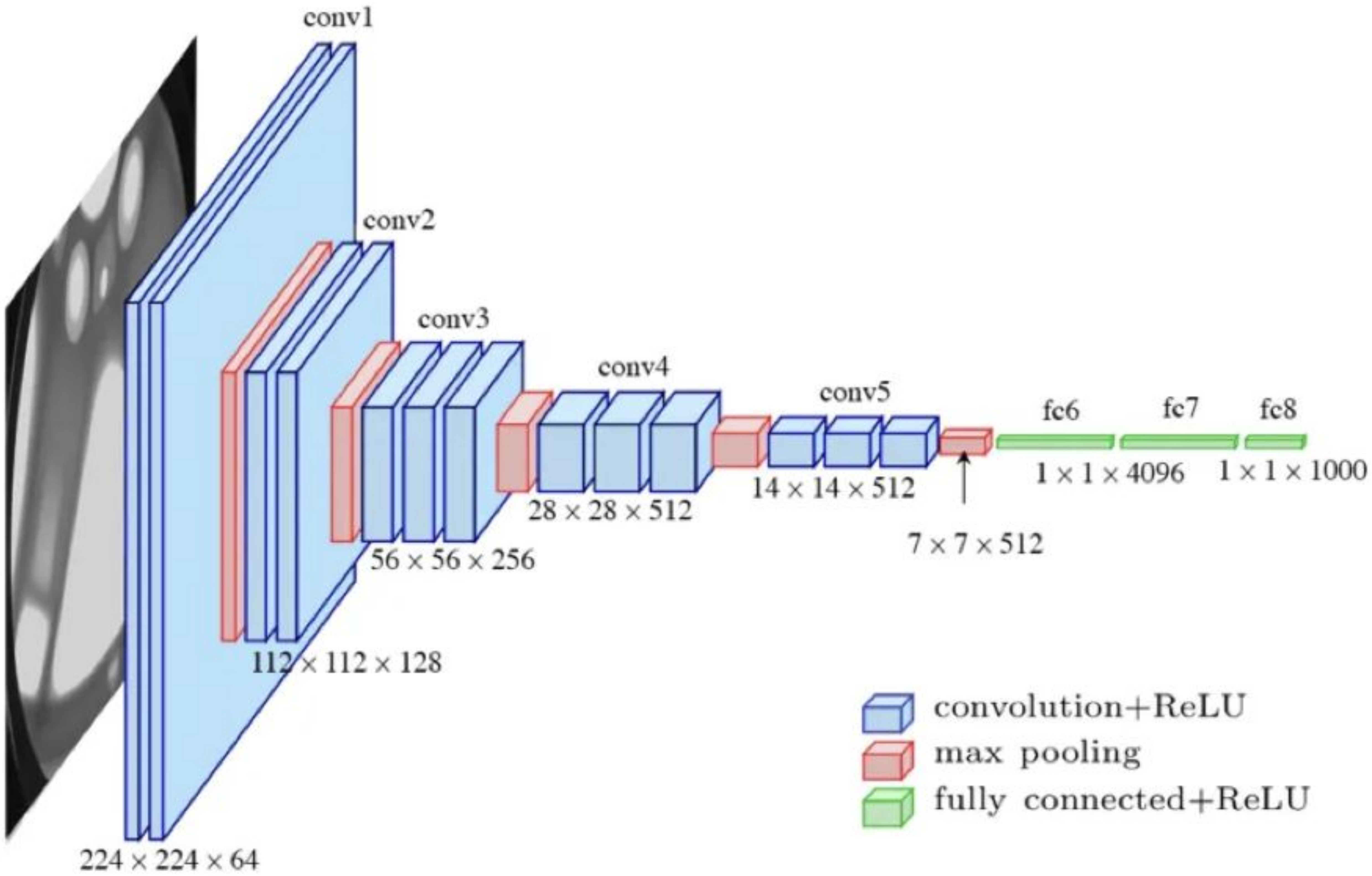
VGG-Net

- It is a typical deep Convolutional Neural Network (CNN) design with numerous layers, and the abbreviation VGG stands for Visual Geometry Group.

- The term “deep” describes the number of layers, with VGG-16 or VGG-19 having 16 or 19 convolutional layers, respectively.The VGGNet, created as a deep neural network, outperforms benchmarks on a variety of tasks and datasets outside of ImageNet.

- Innovative object identification models are built using the VGG architecture.It also remains one of the most often used image recognition architectures today.

VGG-NET ARCHITECTURE



VGG-NET

- **Inputs:** The VGGNet accepts 224x224-pixel images as input. To maintain a consistent input size for the ImageNet competition, the model's developers chopped out the central 224x224 patches in each image

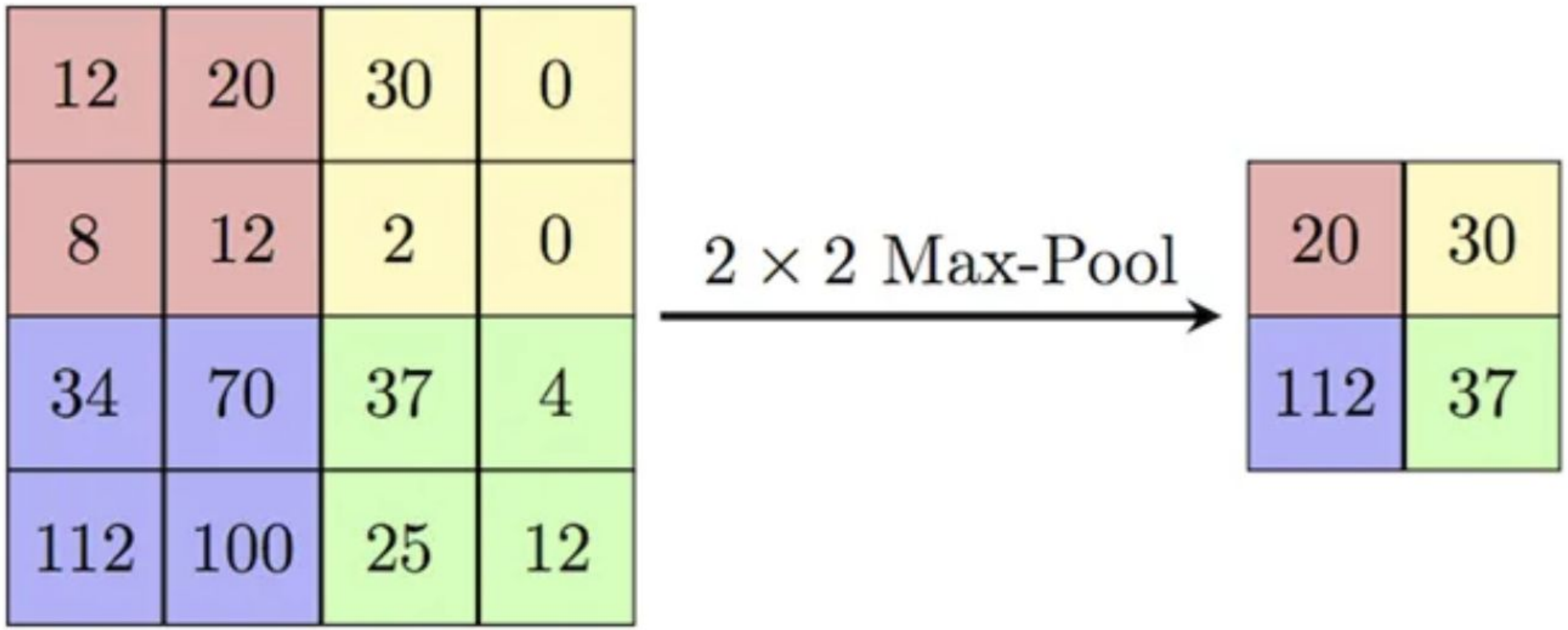
- **Convolutional Layers:** VGG's convolutional layers use the smallest feasible receptive field, to record left-to-right and up-to-down movement. Additionally, 11 convolution filters are used to transform the input linearly.

- The next component is a **ReLU** unit, a significant advancement from AlexNet that shortens training time. Rectified linear unit activation function, or ReLU, is a piecewise linear function that, if the input is positive, outputs the input; otherwise, the output is zero.

- **Hidden Layers:** The VGG network's hidden layers all make use of ReLU. Local Response Normalization (LRN) is typically not used with VGG as it increases memory usage and training time. Furthermore, it doesn't increase overall accuracy.

- **Fully Connected Layers:** The VGGNet contains three layers with full connectivity. The first two levels each have 4096 channels, while the third layer has 1000 channels with one channel for each class.

MAX-POOLING

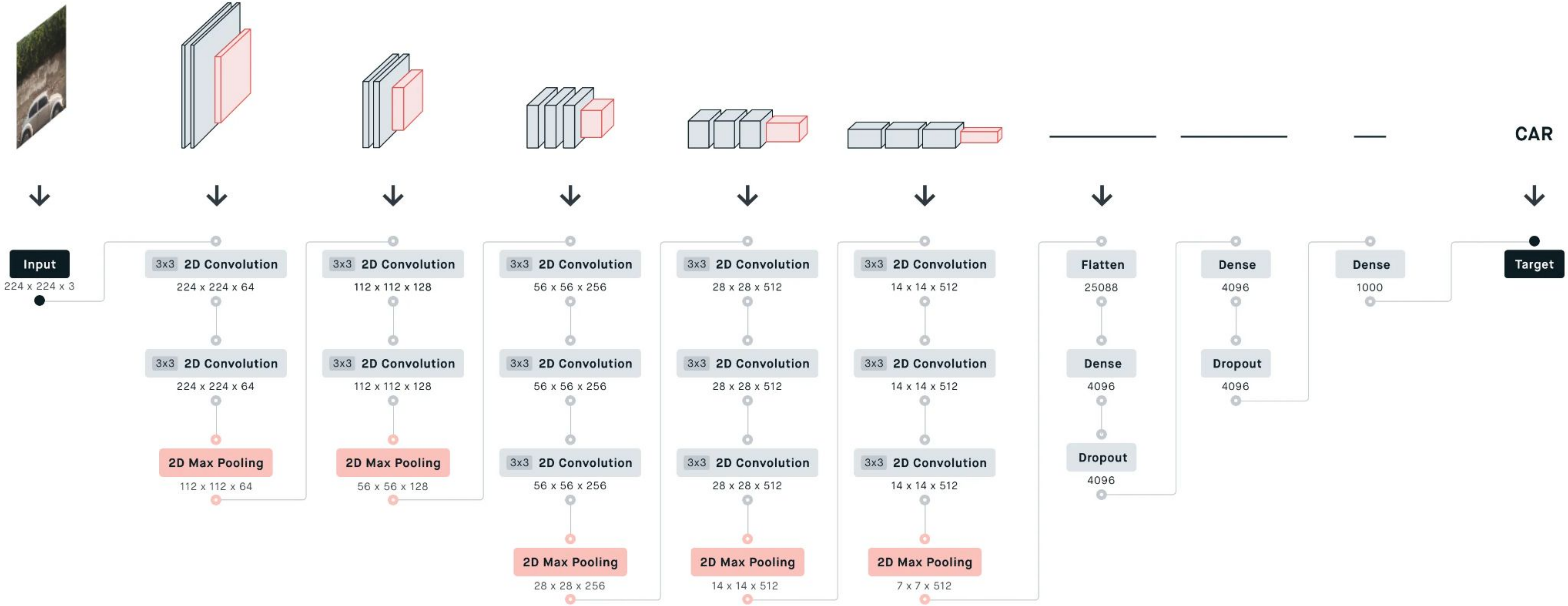


Max Pooling

MAX-POOLING

- The main idea behind a pooling layer is to “accumulate” features from maps generated by convolving a filter over an image.
- Formally, its function is to progressively reduce the spatial size of the representation to reduce the number of parameters and computations in the network. The most common form of pooling is max pooling.
- Max pooling is done in part to help over-fitting by providing an abstracted form of the representation. As well, it reduces the computational cost by reducing the number of parameters to learn and provides basic translation invariance to the internal representation.
- Max pooling is done by applying a max filter to (usually) non-overlapping sub-regions of the initial representation.

UNDERSTANDING VGG-16



Limitations Of VGG 16:

- It is very slow to train (the original VGG model was trained on Nvidia Titan GPU for 2–3 weeks).

- The size of VGG-16 trained imageNet weights is 528 MB. So, it takes quite a lot of disk space and bandwidth which makes it inefficient.

- 138 million parameters lead to exploding gradients problem.

THANK YOU